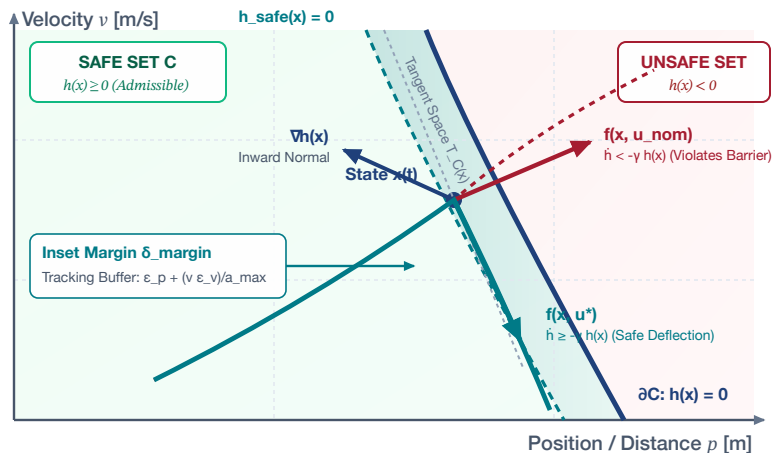


CONTROL BARRIER FUNCTIONS (CBF) & MINIMAL INTERVENTION

Forward Invariance in State Space and Orthogonal Quadratic Program Projection in Control Space

(a) State Space Invariance: Safe Set C & Tangential Filtering



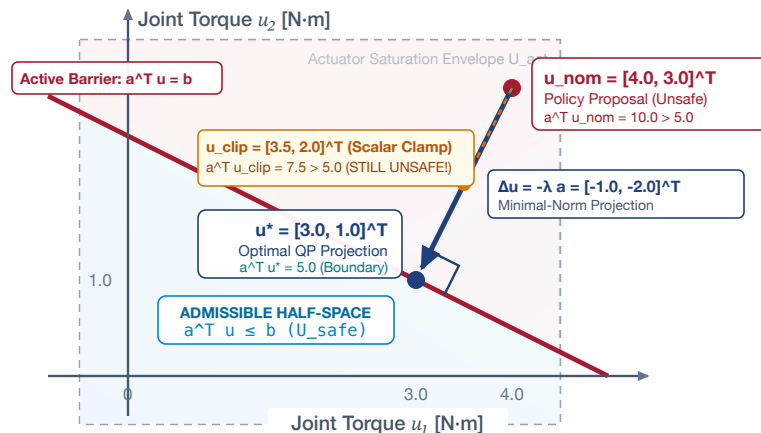
FORWARD INVARIANCE CERTIFICATE

$$\dot{h}(x, u) = L_f h(x) + L_g h(x) u \geq -\gamma h(x)$$

- **Invariant Safe Set:** Zero-superlevel set $C = \{x \in X \mid h(x) \geq 0\}$ is forward invariant.
- **Nagumo's Theorem:** Control vector field $f(x, u^*)$ remains inside tangent cone $T_C(x)$.
- **Tracking Inset Margin:** $h_{\text{safe}}(x) = h(x) - \delta_{\text{margin}}$ buffers tracking error ϵ_{track} .
- **Kinematic Inset Formula:** $\delta_{\text{margin}} = \epsilon_p + (v \epsilon_v)/a_{\text{max}}$ (accounts for loop delay).

Result: Policy commands are filtered tangentially without task abortion.

(b) Control Space: Orthogonal QP Projection onto Admissible Half-Space



QUADRATIC PROGRAM SAFETY FILTER

$$u^* = \arg\min_{u \in U} \frac{1}{2} \|u - u_{\text{nom}}\|^2 \text{ s.t. } a^T u \leq b$$

- **Closed-Form Analytical Solution:** $u^* = u_{\text{nom}} - (\max\{0, a^T u_{\text{nom}} - b\} / \|a\|^2) a$
- **Zero Intervention Interior:** If $a^T u_{\text{nom}} \leq b$, then $u^* = u_{\text{nom}}$ (unmodified policy intent).
- **Coupled Multi-Axis Dynamics:** Preserves Cartesian wrench alignment vs scalar clipping.
- **Policy Health Telemetry:** Rolling norm $\Sigma \|u^* - u_{\text{nom}}\| \Delta t$ tracks upstream model drift.

Guarantee: Microsecond deterministic QP projection on embedded MCU SRAM.